3.26 Hauptidealsatz

The main result of this section is the following result due to Krull

Proposition 3.26.1 (Generalized Hauptidealsatz for Noetherian Rings)
Suppose A is a Noetherian ring then the following properties hold

a) Every prime ideal of height n is minimal over some ideal $\mathfrak{a} := (x_1, ..., x_n)$. Furthermore $\mathfrak{a}$ may be chosen such that $\operatorname{ht}(\mathfrak{a}) = n$ and every minimal prime of $\mathfrak{a}$ is of height n
b) We have the following characterization of height

$$\operatorname{ht}(\mathfrak{p}) = \min\{n \mid \mathfrak{p} \text{ minimal over } (x_1, ..., x_n)\}$$

In particular if $\mathfrak{p}$ is minimal over $(x_1, ..., x_n)$ then $\operatorname{ht}(\mathfrak{p}) \leq n$.

In full generality the proof is quite subtle and in most cases of interest a simpler proof is possible. Therefore we introduce the following notions.

Definition 3.26.2 (Hauptidealsatz Ring)
*We say a ring A is a **generalized hauptidealsatz** ring if for all $x_1, ..., x_n \in A$ and $\mathfrak{p}$ prime ideals minimal over $(x_1, ..., x_n)$ we have $\operatorname{ht}(\mathfrak{p}) \leq n$*

*We say a ring A is simply a **hauptidealsatz** ring if this holds for $n = 1$.*

Lemma 3.26.3
Let A be a hauptidealsatz ring and $\mathfrak{p}$ a prime ideal. Then $A_{\mathfrak{p}}$ is hauptidealsatz.

Proof. Any prime ideal of $A_{\mathfrak{p}}$ has the form $\mathfrak{q}A_{\mathfrak{p}}$ by the correspondence of ideals under localization (3.5.19). If $\mathfrak{q}A_{\mathfrak{p}}$ is minimal over $(f/s) = (f/1)$, then clearly $(f) \subseteq \mathfrak{q}$. Furthermore $(f) \subseteq \mathfrak{q}' \implies (f/1) \subseteq \mathfrak{q}'A_{\mathfrak{p}}$. So $\mathfrak{q}$ is also minimal over (f) and therefore has height at most 1 by assumption. By the same result $\mathfrak{q}A_{\mathfrak{p}}$ has height at most 1. ∎

Proposition 3.26.4 (Hauptidealsatz ⇒ Generalized Hauptidealsatz (Geometric))
Suppose that A is catenary, and hauptidealsatz for every quotient by a prime ideal. Then A is generalized hauptidealsatz, and so is every localization at a prime ideal.

Proof. We consider the case first that A is integral (which means it is hauptidealsatz by assumption because (0) is prime) and assume wlog that $n > 1$. Let $\mathfrak{p}$ be a minimal prime of $(x_1, ..., x_n)$ and $\mathfrak{q}$ a minimal prime of $(x_1, ..., x_{n-1})$. By considering the localization $A_{\mathfrak{p}}$ we may choose $\mathfrak{q} \subseteq \mathfrak{p}$. By induction $\operatorname{ht}(\mathfrak{q}) \leq n - 1$. Then $\mathfrak{p}/\mathfrak{q}$ is a minimal prime over $(x_n + \mathfrak{q})$ and therefore by assumption has height at most 1. By the codimension formula (3.21.9) we see $\operatorname{ht}(\mathfrak{p}) \leq n$ as required.

For the general case, suppose $\mathfrak{p}$ is minimal over $(x_1, ..., x_n)$ and let $\mathfrak{r} \subseteq \mathfrak{p}$ be a minimal prime. Then the ring $A/\mathfrak{r}$ is integral and we see that $\operatorname{ht}(\mathfrak{p}/\mathfrak{r}) \leq n$ by the first part. Taking the supremum over all minimal primes $\mathfrak{r}$ shows that $\operatorname{ht}(\mathfrak{p}) \leq n$.

For the last statement we need only show that every localization at a prime ideal satisfies the hypotheses of the theorem. By correspondence of ideals under localisation (3.5.19) we see that every such ring is catenary. By (3.5.25) $A_{\mathfrak{p}}/\mathfrak{q}A_{\mathfrak{p}} \cong (A/\mathfrak{q})_{\mathfrak{p}/\mathfrak{q}}$ for any prime ideal $\mathfrak{q} \subseteq \mathfrak{p}$ so every quotient by a prime ideal is hauptidealsatz by the previous Lemma. ∎

The catenary assumption may be relaxed by making a more complicated argument.

Proposition 3.26.5 (Hauptidealsatz ⇒ Generalized Hauptidealsatz (Algebraic))
Suppose A is a ring such that every quotient by a prime ideal satisfies the hauptidealsatz. Then A is generalized hauptidealsatz and so is every localization at a prime ideal $\mathfrak{p}$.

Proof. We prove this by induction on n for a fixed ring A. Let $\mathfrak{p}$ be a minimal prime over $(x_1, ..., x_n)$. Suppose $\operatorname{ht}(\mathfrak{p}) > n$ then we may choose a saturated chain $\mathfrak{q} \subsetneq \mathfrak{p}$ such that $\operatorname{ht}(\mathfrak{q}) \geq n$. By minimality of $\mathfrak{p}$ we have $\mathfrak{q} \cap \{x_1, ..., x_n\} = \{x_1, ..., x_r\}$ with $r < n$ after a suitable reordering (possibly with $r = 0$). Furthermore we may choose $\mathfrak{q}$ such that r is maximal. Then by the induction hypothesis $\mathfrak{q}$ is not minimal over $(x_1, ..., x_r)$ and there is a chain

$$(x_1, ..., x_r) \subseteq \mathfrak{r} \subsetneq \mathfrak{q} \subsetneq \mathfrak{p}$$

We claim that $\mathfrak{p}$ is minimal over $(\mathfrak{r}, x_{r+1})$ for suppose

$$(\mathfrak{r}, x_{r+1}) \subseteq \mathfrak{p}' \subseteq \mathfrak{p}$$

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then by construction of $\mathfrak{q}$ (r was maximal) we see that $\mathfrak{p}' = \mathfrak{p}$. Taking quotients by $\mathfrak{r}$ the hauptidealsatz property shows that $\mathrm{ht}(\mathfrak{p}/\mathfrak{r}) \leq 1$. On the other hand we have a chain

$$(0) \subsetneq \mathfrak{q}/\mathfrak{r} \subsetneq \mathfrak{p}/\mathfrak{r}$$

which is a contradiction.

The final statement follows a similar argument as before. ∎

Remark 3.26.6
This argument is from [Kap74] (Sec. 3.2 Ex. 6).

Before proceeding to the main result we need a some technical results related to height of ideals.

Lemma 3.26.7
Let $\mathfrak{q} \subseteq \mathfrak{p}$ be prime ideals then

$$\mathrm{ht}(\mathfrak{q}) \leq \mathrm{ht}(\mathfrak{p})$$

with equality iff $\mathfrak{p} = \mathfrak{q}$.

Proof. The inequality is clear since any maximal chain below $\mathfrak{q}$ may be extended to a maximal chain below $\mathfrak{p}$. Similarly $\mathfrak{q} \subsetneq \mathfrak{p}$ implies that the inequality is strict. ∎

Lemma 3.26.8
Suppose $\mathfrak{a} \subseteq \mathfrak{b}$. Then

$$\mathrm{ht}(\mathfrak{a}) \leq \mathrm{ht}(\mathfrak{b})$$

If these are equal and $\mathfrak{b}$ is prime, then it must be a minimal prime of $\mathfrak{a}$.

Proof. We consider first the case $\mathfrak{b} = \mathfrak{p}$ is prime. By (3.3.44) there is a minimal prime $\mathfrak{p}'$ such that $\mathfrak{a} \subseteq \mathfrak{p}' \subseteq \mathfrak{p}$. By definition $\mathrm{ht}(\mathfrak{a}) \leq \mathrm{ht}(\mathfrak{p}')$. Furthermore by the previous Lemma $\mathrm{ht}(\mathfrak{p}') \leq \mathrm{ht}(\mathfrak{p})$, which yields the required result.

Suppose $\mathrm{ht}(\mathfrak{p}) = \mathrm{ht}(\mathfrak{a})$ then by definition $\mathrm{ht}(\mathfrak{a}) \leq \mathrm{ht}(\mathfrak{p}') \leq \mathrm{ht}(\mathfrak{p})$ and therefore we conclude $\mathrm{ht}(\mathfrak{p}) = \mathrm{ht}(\mathfrak{p}')$. By the previous lemma we conclude that $\mathfrak{p} = \mathfrak{p}'$ is a minimal prime.

For the general case every minimal prime of $\mathfrak{b}$ also contains $\mathfrak{a}$ so the inequality follows by taking the minimum over all minimal primes. ∎

Lemma 3.26.9
Let $\mathfrak{a} \subseteq \mathfrak{b}$ be ideals such that there exists $x \in \mathfrak{b}$ not contained in any minimal prime of $\mathfrak{a}$. Then

$$\mathrm{ht}(\mathfrak{a}) < \mathrm{ht}(\mathfrak{b})$$

Proof. By the previous Lemma we have $\mathrm{ht}(\mathfrak{a}) \leq \mathrm{ht}(\mathfrak{b})$. Suppose they are equal then there is a minimal prime $\mathfrak{p}$ of $\mathfrak{b}$ of the same height, which by (3.26.8) is a minimal prime of $\mathfrak{a}$. This must contain x which then contradicts the assumption. ∎

Proposition 3.26.10 (Prime Avoidance Theorem)
Let $\mathfrak{p}_1, ..., \mathfrak{p}_n$ be ideals and $\mathfrak{a}$ an ideal such that $\mathfrak{a} \nsubseteq \mathfrak{p}_i$ for all $i = 1...n$. Then there exists $x \in \mathfrak{a} \setminus (\mathfrak{p}_1 \cup ... \cup \mathfrak{p}_n)$.

Proof. We proceed by induction on the number of prime ideals n, the case $n = 1$ being trivial. Consider then the case of $n+1$ prime ideals and suppose on the contrary that $\mathfrak{a} \subseteq \mathfrak{p}_1 \cup ... \cup \mathfrak{p}_{n+1}$. By the induction hypothesis there exists $x_i \in \mathfrak{a} \setminus \bigcup_{j \neq i} \mathfrak{p}_j$, and therefore by our assumption $x_i \in (\mathfrak{a} \cap \mathfrak{p}_i) \setminus \bigcup_{j \neq i} \mathfrak{p}_j$. Define

$$y := x_1 ... x_n$$

Then $y \notin \mathfrak{p}_{n+1}$ by primality and clearly $y \in \mathfrak{a} \cap \mathfrak{p}_1 \cap ... \cap \mathfrak{p}_n$. Define $z := y + x_{n+1}$ then we see

a) $z \in \mathfrak{a}$
b) $z \notin \mathfrak{p}_{n+1}$
c) $z \notin \mathfrak{p}_i$ for $i = 1...n$

which contradicts our original assumption. ∎

Lemma 3.26.11
*Let A be a **generalized hauptidealsatz** ring and $\mathfrak{a} = (x_1, ..., x_n)$. Then the following are equivalent*

a) $\mathrm{ht}(\mathfrak{a}) = n$
b) every minimal prime of $\mathfrak{a}$ has height n

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Proof. a) $\implies$ b) By definition this means every minimal prime has height at least n, and the reverse inequality follows from generalized hauptidealsatz assumption.

b) $\implies$ a) Recall from (3.21.1) we may take the infimum over only minimal prime ideals. ∎

Proposition 3.26.12 (Alternative characterization of height)
*Suppose A is a Noetherian ring satisfying **generalized hauptidealsatz**, then the following properties hold*

a) Every prime ideal of height n is minimal over some ideal $\mathfrak{a} := (x_1, ..., x_n)$. Furthermore $\mathfrak{a}$ may be chosen such that $\operatorname{ht}(\mathfrak{a}) = n$ and every minimal prime is of height n
b) We have the following characterization of height

$$\operatorname{ht}(\mathfrak{p}) = \min\{n \mid \mathfrak{p} \text{ minimal over } (x_1, ..., x_n)\}$$

In particular every prime ideal has finite height.

Proof. We first demonstrate a). Suppose $\operatorname{ht}(\mathfrak{p}) = n$ then there is a saturated chain of prime ideals

$$\mathfrak{p}_0 \subsetneq ... \subsetneq \mathfrak{p}_n = \mathfrak{p}$$

It's clear from the above chain that $\operatorname{ht}(\mathfrak{p}_i) \geq i$. Furthermore by (3.26.7) $\operatorname{ht}(\mathfrak{p}_i) < \operatorname{ht}(\mathfrak{p}_{i+1})$ whence we see $\operatorname{ht}(\mathfrak{p}_i) = i$.

We show by induction that there are elements $x_1, ..., x_n \in A$ for which $\mathfrak{p}_i$ is a minimal prime over $\mathfrak{a}_i := (x_1, ..., x_i)$ for all $i = 0...n$ and $\operatorname{ht}(\mathfrak{a}_i) = i$.

The case $i = 0$ is clear, so suppose $i > 0$. In general by (3.21.6) there are finitely many minimal primes over $\mathfrak{a}_{i-1}$ say $\mathfrak{q}_1, ..., \mathfrak{q}_r$ which all have height $i - 1$ by (3.26.11). In particular $\mathfrak{p}_i \not\subseteq \mathfrak{q}_k$.

By prime avoidance (3.26.10) we may choose $x_i \in \mathfrak{p}_i \setminus (\mathfrak{q}_1 \cup ... \cup \mathfrak{q}_r)$. Clearly

$$\mathfrak{a}_{i-1} \subseteq \mathfrak{a}_i \subseteq \mathfrak{p}_i$$

Then by (3.26.8) and (3.26.9) we have $i - 1 = \operatorname{ht}(\mathfrak{a}_{i-1}) < \operatorname{ht}(\mathfrak{a}_i) \leq i$ whence $\operatorname{ht}(\mathfrak{a}_i) = i$. By (3.26.8) again we see $\mathfrak{p}_i$ is a minimal prime of $\mathfrak{a}_i$. This completes the inductive step.

To show b) let m denote the right hand side. Then by the generalized hauptidealsatz hypothesis $\operatorname{ht}(\mathfrak{p}) \leq m$. On the other hand by part a) we have $m \leq \operatorname{ht}(\mathfrak{p})$ whence they are equal.

Finally by the Noetherian hypothesis $\mathfrak{p}$ is finitely generated and clearly minimal over itself. ∎

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